

SYNTHESIS REPORTS

SMOOTHENING

POWER REPORT:

Power Report

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1. Summary

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+-----+
| Total On-Chip Power (W) | 95.871 (Junction temp exceeded!) |
| Design Power Budget (W) | Unspecified*                |
| Power Budget Margin (W) | NA                    |
| Dynamic (W)             | 95.387                |
| Device Static (W)       | 0.485                  |
| Effective TJA (C/W)     | 5.0                    |
| Max Ambient (C)         | 0.0                    |
| Junction Temperature (C) | 125.0                  |
| Confidence Level        | Low                    |
| Setting File            | ---                    |
| Simulation Activity File | ---                    |
| Design Nets Matched     | NA                      |
+-----+
```

* Specify Design Power Budget using, set_operating_conditions -design_power_budget <value in Watts>

1.1 On-Chip Components

+-----+-----+-----+-----+-----+				
On-Chip	Power (W)	Used	Available	Utilization (%)
+-----+-----+-----+-----+-----+				
Slice Logic	11.774	31763	---	---
LUT as Logic	10.273	10825	20800	52.04
Register	0.778	16528	41600	39.73
F7/F8 Muxes	0.553	3942	32600	12.09
CARRY4	0.166	34	8150	0.42
BUFG	0.006	1	32	3.13
Others	0.000	2	---	---
Signals	26.027	32315	---	---
I/O	57.580	16388	106	15460.38
Static Power	0.485			
Total	95.870			
+-----+-----+-----+-----+-----+				

1.2 Power Supply Summary

+-----+-----+-----+-----+-----+				
Source	Voltage (V)	Total (A)	Dynamic (A)	Static (A)
+-----+-----+-----+-----+-----+				
Vccint	1.000	71.176	70.835	0.341
Vccaux	1.800	0.079	0.025	0.053
Vcco33	3.300	0.000	0.000	0.000
Vcco25	2.500	0.000	0.000	0.000
Vcco18	1.800	13.616	13.615	0.001
Vcco15	1.500	0.000	0.000	0.000
Vcco135	1.350	0.000	0.000	0.000
Vcco12	1.200	0.000	0.000	0.000
Vccaux_io	1.800	0.000	0.000	0.000
Vccbram	1.000	0.010	0.000	0.010
MGTAVcc	1.000	0.000	0.000	0.000
MGTAVtt	1.200	0.000	0.000	0.000
Vccadc	1.800	0.020	0.000	0.020
+-----+-----+-----+-----+-----+				

1.3 Confidence Level

User Input Data	Confidence	Details	Action
Design implementation state Low Design is synthesized Accuracy of the tool is not optimal until design is fully placed and routed Clock nodes activity Low User specified less than 75% of clocks Provide missing clock activity with a constraint file, simulation results or by editing the "By Clock Domain" view I/O nodes activity Low More than 75% of inputs are missing user specification Provide missing input activity with simulation results or by editing the "By Resource Type -> I/Os" view Internal nodes activity Medium User specified less than 25% of internal nodes Provide missing internal nodes activity with simulation results or by editing the "By Resource Type" views Device models High Device models are Production Overall confidence level Low			

2. Settings

2.1 Environment

Ambient Temp (C)	25.0
ThetaJA (C/W)	5.0
Airflow (LFM)	250
Heat Sink	medium (Medium Profile)
ThetaSA (C/W)	4.6
Board Selection	medium (10"x10")
# of Board Layers	12to15 (12 to 15 Layers)
Board Temperature (C)	25.0

2.2 Clock Constraints

+-----+	+-----+	+-----+
Clock	Domain	Constraint (ns)
+-----+	+-----+	+-----+

3. Detailed Reports

3.1 By Hierarchy

+-----+	+-----+
Name	Power (W)
+-----+	+-----+
target_module	95.385
+-----+	+-----+

1 Infos, 1 Warnings, 0 Critical Warnings and 0 Errors encountered.
report_power completed successfully
report_power: Time (s): cpu = 00:00:43 ; elapsed = 00:00:22 . Memory (MB): peak = 7702.156 ;
gain = 4.719 ; free physical = 297077 ; free virtual = 563027

TIMING REPORT:

Timing Report

Slack: inf
Source: padded_image_reg[23][7][0]/C
(rising edge-triggered cell FDRE)
Destination: blurred_image_reg[0][10][5]/D
Path Group: (none)
Path Type: Max at Slow Process Corner
Data Path Delay: 15.258ns (logic 6.886ns (45.130%) route 8.372ns (54.870%))
Logic Levels: 21 (CARRY4=8 FDRE=1 LUT2=1 LUT3=1 LUT5=1 LUT6=6 MUXF7=2 MUXF8=1)

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
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```

-----
      FDRE          0.000  0.000 r padded_image_reg[23][7][0]/C
      FDRE (Prop_fdre_C_Q)      0.456  0.456 r padded_image_reg[23][7][0]/Q
      net (fo=3, unplaced)      0.983  1.439 padded_image_reg[23][7]__0[0]
      LUT6 (Prop_lut6_I0_O)      0.295  1.734 r
blurred_image[13][14][7]_i_5613/O
      net (fo=1, unplaced)      0.000  1.734 blurred_image[13][14][7]_i_5613_n_0
      MUXF7 (Prop_muxf7_I1_O)    0.245  1.979 r
blurred_image_reg[13][14][7]_i_4352/O
      net (fo=1, unplaced)      0.000  1.979
blurred_image_reg[13][14][7]_i_4352_n_0
      MUXF8 (Prop_muxf8_I0_O)    0.104  2.083 r
blurred_image_reg[13][14][7]_i_3286/O
      net (fo=1, unplaced)      0.611  2.694
blurred_image_reg[13][14][7]_i_3286_n_0
      LUT5 (Prop_lut5_I2_O)      0.319  3.013 r
blurred_image[13][14][7]_i_2544/O
      net (fo=3, unplaced)      0.920  3.933 blurred_image[13][14][7]_i_2544_n_0
      LUT6 (Prop_lut6_I1_O)      0.124  4.057 r
blurred_image[13][14][7]_i_1804/O
      net (fo=1, unplaced)      0.000  4.057 blurred_image[13][14][7]_i_1804_n_0
      MUXF7 (Prop_muxf7_I1_O)    0.245  4.302 r
blurred_image_reg[13][14][7]_i_1152/O
      net (fo=1, unplaced)      0.905  5.207
blurred_image_reg[13][14][7]_i_1152_n_0
      LUT6 (Prop_lut6_I0_O)      0.298  5.505 r blurred_image[13][14][7]_i_645/O
      net (fo=2, unplaced)      0.650  6.155 blurred_image[13][14][7]_i_645_n_0
      CARRY4 (Prop_carry4_DI[1]_O[2])
          0.554  6.709 r blurred_image_reg[13][14][7]_i_315/O[2]
      net (fo=2, unplaced)      0.653  7.362
blurred_image_reg[13][14][7]_i_315_n_5
      CARRY4 (Prop_carry4_DI[1]_O[2])
          0.731  8.093 r blurred_image_reg[13][14][7]_i_143/O[2]
      net (fo=2, unplaced)      0.463  8.556
blurred_image_reg[13][14][7]_i_143_n_5
      LUT6 (Prop_lut6_I5_O)      0.301  8.857 r blurred_image[13][14][7]_i_312/O
      net (fo=1, unplaced)      0.000  8.857 blurred_image[13][14][7]_i_312_n_0
      CARRY4 (Prop_carry4_S[3]_O[3])
          0.248  9.105 r blurred_image_reg[13][14][7]_i_142/O[3]
      net (fo=1, unplaced)      0.618  9.723 C[3]
      LUT2 (Prop_lut2_I1_O)      0.307  10.030 r blurred_image[13][14][7]_i_116/O
      net (fo=1, unplaced)      0.000  10.030 blurred_image[13][14][7]_i_116_n_0
      CARRY4 (Prop_carry4_S[3]_CO[3])
          0.401  10.431 r blurred_image_reg[13][14][7]_i_53/CO[3]

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net (fo=1, unplaced)      0.009  10.440
blurred_image_reg[13][14][7]_i_53_n_0
  CARRY4 (Prop_carry4_CI_O[0])
    0.235  10.675 r blurred_image_reg[13][14][7]_i_30/O[0]
net (fo=2, unplaced)      0.322  10.997
blurred_image_reg[13][14][7]_i_30_n_7
  LUT3 (Prop_lut3_I1_O)    0.295  11.292 r blurred_image[13][14][7]_i_24/O
net (fo=2, unplaced)      0.650  11.942 blurred_image[13][14][7]_i_24_n_0
  CARRY4 (Prop_carry4_DI[1]_O[2])
    0.554  12.496 r blurred_image_reg[13][14][7]_i_15/O[2]
net (fo=2, unplaced)      0.463  12.959 p_0_in[6]
  LUT6 (Prop_lut6_I1_O)    0.301  13.260 r blurred_image[13][14][3]_i_3/O
net (fo=2, unplaced)      0.460  13.720 blurred_image[13][14][3]_i_3_n_0
  LUT6 (Prop_lut6_I0_O)    0.124  13.844 r blurred_image[13][14][3]_i_7/O
net (fo=1, unplaced)      0.000  13.844 blurred_image[13][14][3]_i_7_n_0
  CARRY4 (Prop_carry4_S[3]_CO[3])
    0.401  14.245 r blurred_image_reg[13][14][3]_i_1/CO[3]
net (fo=1, unplaced)      0.000  14.245
blurred_image_reg[13][14][3]_i_1_n_0
  CARRY4 (Prop_carry4_CI_O[1])
    0.348  14.593 r blurred_image_reg[13][14][7]_i_2/O[1]
net (fo=1024, unplaced)    0.665  15.258
blurred_image_reg[13][14][7]_i_2_n_6
  FDRE
    r blurred_image_reg[0][10][5]/D

```

UTILIZATION REPORT:

Utilization Design Information

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1. Slice Logic

Site Type	Used	Fixed	Available	Util%
Slice LUTs*	10825	0	20800	52.04
LUT as Logic	10825	0	20800	52.04
LUT as Memory	0	0	9600	0.00
Slice Registers	16528	0	41600	39.73
Register as Flip Flop	16528	0	41600	39.73
Register as Latch	0	0	41600	0.00
F7 Muxes	2866	0	16300	17.58
F8 Muxes	1076	0	8150	13.20

* Warning! The Final LUT count, after physical optimizations and full implementation, is typically lower. Run opt_design after synthesis, if not already completed, for a more realistic count.

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	-	-	-
0	-	-	Set
0	-	-	Reset
0	-	Set	-
0	-	Reset	-
0	Yes	-	-
14	Yes	-	Set
130	Yes	-	Reset
0	Yes	Set	-
16384	Yes	Reset	-

2. Memory

Site Type	Used	Fixed	Available	Util%
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	Used	Fixed	Available	Util%
Block RAM Tile	0	0	50	0.00
RAMB36/FIFO*	0	0	50	0.00
RAMB18	0	0	100	0.00

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

3. DSP

	Used	Fixed	Available	Util%
Site Type	Used	Fixed	Available	Util%
DSPs	0	0	90	0.00

4. IO and GT Specific

	Used	Fixed	Available	Util%
Site Type	Used	Fixed	Available	Util%
Bonded IOB	16388	0	106	15460.38
Bonded IPADs	0	0	10	0.00
Bonded OPADs	0	0	4	0.00
PHY_CONTROL	0	0	5	0.00
PHASER_REF	0	0	5	0.00
OUT_FIFO	0	0	20	0.00
IN_FIFO	0	0	20	0.00
IDELAYCTRL	0	0	5	0.00
IBUFDS	0	0	104	0.00
GTPE2_CHANNEL	0	0	2	0.00
PHASER_OUT/PHASER_OUT_PHY	0	0	20	0.00
PHASER_IN/PHASER_IN_PHY	0	0	20	0.00
IDELAYE2/IDELAYE2_FINEDELAY	0	0	250	0.00
IBUFDS_GTE2	0	0	2	0.00
ILOGIC	0	0	106	0.00
OLOGIC	0	0	106	0.00

5. Clocking

Site Type	Used	Fixed	Available	Util%
BUFGCTRL	1	0	32	3.13
BUFIO	0	0	20	0.00
MMCM2_ADV	0	0	5	0.00
PLLE2_ADV	0	0	5	0.00
BUFMRC	0	0	10	0.00
BUFHCE	0	0	72	0.00
BUFR	0	0	20	0.00

6. Specific Feature

Site Type	Used	Fixed	Available	Util%
BSCANE2	0	0	4	0.00
CAPTUREE2	0	0	1	0.00
DNA_PORT	0	0	1	0.00
EFUSE_USR	0	0	1	0.00
FRAME_ECCE2	0	0	1	0.00
ICAPE2	0	0	2	0.00
PCIE_2_1	0	0	1	0.00
STARTUPE2	0	0	1	0.00
XADC	0	0	1	0.00

7. Primitives

Ref Name	Used	Functional Category
FDRE	16384	Flop & Latch
LUT6	9256	LUT
IBUF	8195	IO

OBUF	8193		IO	
MUXF7	2866		MuxFx	
MUXF8	1076		MuxFx	
LUT5	942		LUT	
LUT3	712		LUT	
LUT4	207		LUT	
LUT2	138		LUT	
FDCE	130		Flop & Latch	
CARRY4	34		CarryLogic	
FDPE	14		Flop & Latch	
LUT1	1		LUT	
BUFG	1		Clock	
+-----+-----+-----+				

8. Black Boxes

+-----+-----+				
Ref Name	Used			
+-----+-----+				

9. Instantiated Netlists

+-----+-----+				
Ref Name	Used			
+-----+-----+				

report_utilization: Time (s): cpu = 00:00:00.62 ; elapsed = 00:00:00.61 . Memory (MB): peak = 7702.156 ; gain = 0.000 ; free physical = 296827 ; free virtual = 562777

SHARPENING

POWER REPORT:

Power Report

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1. Summary

```
+-----+-----+
| Total On-Chip Power (W) | 96.476 (Junction temp exceeded!) |
| Design Power Budget (W) | Unspecified* |
| Power Budget Margin (W) | NA |
| Dynamic (W) | 95.991 |
| Device Static (W) | 0.485 |
| Effective TJA (C/W) | 5.0 |
| Max Ambient (C) | 0.0 |
| Junction Temperature (C) | 125.0 |
| Confidence Level | Low |
| Setting File | --- |
| Simulation Activity File | --- |
| Design Nets Matched | NA |
+-----+-----+
```

* Specify Design Power Budget using, set_operating_conditions -design_power_budget <value in Watts>

1.1 On-Chip Components

```
+-----+-----+-----+-----+-----+
```

On-Chip	Power (W)	Used	Available	Utilization (%)
Slice Logic	7.639	29281	---	---
LUT as Logic	6.984	8053	20800	38.72
F7/F8 Muxes	0.514	3116	32600	9.56
CARRY4	0.126	114	8150	1.40
Register	0.010	17846	41600	42.90
BUFG	0.006	1	32	3.13
Others	0.000	2	---	---
Signals	30.574	31087	---	---
I/O	57.768	16457	106	15525.47
Static Power	0.485			
Total	96.465			

1.2 Power Supply Summary

Source	Voltage (V)	Total (A)	Dynamic (A)	Static (A)
Vccint	1.000	71.878	71.537	0.341
Vccaux	1.800	0.053	0.000	0.053
Vcco33	3.300	0.000	0.000	0.000
Vcco25	2.500	0.000	0.000	0.000
Vcco18	1.800	13.587	13.586	0.001
Vcco15	1.500	0.000	0.000	0.000
Vcco135	1.350	0.000	0.000	0.000
Vcco12	1.200	0.000	0.000	0.000
Vccaux_io	1.800	0.000	0.000	0.000
Vccbram	1.000	0.010	0.000	0.010
MGTAVcc	1.000	0.000	0.000	0.000
MGTAVtt	1.200	0.000	0.000	0.000
Vccadc	1.800	0.020	0.000	0.020

1.3 Confidence Level

User Input Data	Confidence	Details	Action
-----+			
Design implementation state	Low	Design is synthesized	Accuracy
of the tool is not optimal until design is fully placed and routed			
Clock nodes activity	Low	User specified less than 75% of clocks	Provide
missing clock activity with a constraint file, simulation results or by editing the "By Clock Domain" view			
I/O nodes activity	Low	More than 75% of inputs are missing user specification	Provide
missing input activity with simulation results or by editing the "By Resource Type -> I/Os" view			
Internal nodes activity	Medium	User specified less than 25% of internal nodes	Provide
missing internal nodes activity with simulation results or by editing the "By Resource Type" views			
Device models	High	Device models are Production	
-----+			
Overall confidence level	Low		
-----+			

2. Settings

2.1 Environment

-----+			
Ambient Temp (C)	25.0		
ThetaJA (C/W)	5.0		
Airflow (LFM)	250		
Heat Sink	medium (Medium Profile)		
ThetaSA (C/W)	4.6		
Board Selection	medium (10"x10")		
# of Board Layers	12to15 (12 to 15 Layers)		
Board Temperature (C)	25.0		
-----+			

2.2 Clock Constraints

3. Detailed Reports

3.1 By Hierarchy

1 Infos, 1 Warnings, 0 Critical Warnings and 0 Errors encountered.
report_power completed successfully
report_power: Time (s): cpu = 00:00:50 ; elapsed = 00:00:16 . Memory (MB): peak = 8235.414 ;
gain = 0.000 ; free physical = 295595 ; free virtual = 561540

TIMING REPORT:

Timing Report

Slack: inf
Source: k_reg[1]/C
(rising edge-triggered cell FDRE)
Destination: multiplication_reg[0][1][7]/D
Path Group: (none)
Path Type: Max at Slow Process Corner
Data Path Delay: 12.771ns (logic 5.029ns (39.378%) route 7.742ns (60.622%))
Logic Levels: 19 (CARRY4=6 FDRE=1 LUT2=2 LUT5=2 LUT6=7 MUXF7=1)

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)
	FDRE	0.000	0.000	r k_reg[1]/C
	FDRE (Prop_fdre_C_Q)	0.456	0.456	r k_reg[1]/Q
	net (fo=20, unplaced)	0.530	0.986	k_reg[1]
	LUT2 (Prop_lut2_I0_O)	0.295	1.281	r multiplication[0][0][7]_i_238/O

```

net (fo=1, unplaced)      0.000  1.281  multiplication[0][0][7]_i_238_n_0
CARRY4 (Prop_carry4_S[1]_CO[3])
      0.550  1.831 r multiplication_reg[0][0][7]_i_132/CO[3]
net (fo=1, unplaced)      0.009  1.840  multiplication_reg[0][0][7]_i_132_n_0
CARRY4 (Prop_carry4_CI_O[1])
      0.348  2.188 r multiplication_reg[0][0][7]_i_71/O[1]
net (fo=306, unplaced)    0.739  2.927  multiplication_reg[0][0][7]_i_71_n_6
LUT6 (Prop_lut6_I1_O)     0.306  3.233 f multiplication[0][0][7]_i_1042/O
net (fo=32, unplaced)     0.973  4.206  multiplication[0][0][7]_i_1042_n_0
LUT6 (Prop_lut6_I0_O)     0.124  4.330 f multiplication[0][0][2]_i_566/O
net (fo=1, unplaced)     1.111  5.441  multiplication[0][0][2]_i_566_n_0
LUT6 (Prop_lut6_I1_O)     0.124  5.565 f multiplication[0][0][2]_i_319/O
net (fo=1, unplaced)     0.449  6.014  multiplication[0][0][2]_i_319_n_0
LUT5 (Prop_lut5_I4_O)     0.124  6.138 f multiplication[0][0][2]_i_188/O
net (fo=1, unplaced)     0.449  6.587  multiplication[0][0][2]_i_188_n_0
LUT6 (Prop_lut6_I1_O)     0.124  6.711 f multiplication[0][0][2]_i_106/O
net (fo=1, unplaced)     0.000  6.711  multiplication[0][0][2]_i_106_n_0
MUXF7 (Prop_muxf7_I1_O)  0.217  6.928 f
multiplication_reg[0][0][2]_i_68/O
net (fo=2, unplaced)     0.916  7.844  multiplication_reg[0][0][2]_i_68_n_0
LUT6 (Prop_lut6_I0_O)     0.298  8.142 f multiplication[0][0][2]_i_38/O
net (fo=5, unplaced)     0.477  8.619  multiplication[0][0][2]_i_38_n_0
LUT6 (Prop_lut6_I0_O)     0.124  8.743 r multiplication[0][0][2]_i_15/O
net (fo=2, unplaced)     0.913  9.656  multiplication[0][0][2]_i_15_n_0
LUT6 (Prop_lut6_I1_O)     0.124  9.780 r multiplication[0][0][2]_i_5/O
net (fo=1, unplaced)     0.000  9.780  multiplication[0][0][2]_i_5_n_0
CARRY4 (Prop_carry4_S[3]_CO[3])
      0.401  10.181 r multiplication_reg[0][0][2]_i_1/CO[3]
net (fo=1, unplaced)     0.009  10.190  multiplication_reg[0][0][2]_i_1_n_0
CARRY4 (Prop_carry4_CI_O[1])
      0.348  10.538 r multiplication_reg[0][0][7]_i_20/O[1]
net (fo=2, unplaced)     0.323  10.861  multiplication_reg[0][0][7]_i_20_n_6
LUT2 (Prop_lut2_I1_O)     0.306  11.167 r multiplication[0][0][6]_i_2/O
net (fo=2, unplaced)     0.460  11.627  multiplication[0][0][6]_i_2_n_0
LUT5 (Prop_lut5_I0_O)     0.124  11.751 r multiplication[0][0][6]_i_5/O
net (fo=1, unplaced)     0.000  11.751  multiplication[0][0][6]_i_5_n_0
CARRY4 (Prop_carry4_S[3]_CO[3])
      0.401  12.152 r multiplication_reg[0][0][6]_i_1/CO[3]
net (fo=1, unplaced)     0.009  12.161  multiplication_reg[0][0][6]_i_1_n_0
CARRY4 (Prop_carry4_CI_O[0])
      0.235  12.396 r multiplication_reg[0][0][7]_i_2/O[0]
net (fo=9, unplaced)     0.375  12.771  multiplication_reg[0][0][7]_i_2_n_7
FDRE
      r multiplication_reg[0][1][7]/D
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UTILIZATION REPORT:

Utilization Design Information

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1. Slice Logic

Site Type	Used	Fixed	Available	Util%
Slice LUTs*	8053	0	20800	38.72
LUT as Logic	8053	0	20800	38.72
LUT as Memory	0	0	9600	0.00
Slice Registers	17846	0	41600	42.90
Register as Flip Flop	17846	0	41600	42.90
Register as Latch	0	0	41600	0.00
F7 Muxes	2080	0	16300	12.76
F8 Muxes	1036	0	8150	12.71

* Warning! The Final LUT count, after physical optimizations and full implementation, is typically lower. Run opt_design after synthesis, if not already completed, for a more realistic count.

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	_	-	-

0	_	-	Set
0	_	-	Reset
0	_	Set	-
0	_	Reset	-
0	Yes	-	-
0	Yes	-	Set
0	Yes	-	Reset
69	Yes	Set	-
17777	Yes	Reset	-
+-----+-----+-----+-----+			

2. Memory

+-----+-----+-----+-----+				
Site Type	Used	Fixed	Available	Util%
+-----+-----+-----+-----+				
Block RAM Tile	0	0	50	0.00
RAMB36/FIFO*	0	0	50	0.00
RAMB18	0	0	100	0.00
+-----+-----+-----+-----+				

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

3. DSP

+-----+-----+-----+-----+				
Site Type	Used	Fixed	Available	Util%
+-----+-----+-----+-----+				
DSPs	0	0	90	0.00
+-----+-----+-----+-----+				

4. IO and GT Specific

+-----+-----+-----+-----+				
Site Type	Used	Fixed	Available	Util%
+-----+-----+-----+-----+				
Bonded IOB	16457	0	106	15525.47

Bonded IPADs		0		0		10		0.00	
Bonded OPADs		0		0		4		0.00	
PHY_CONTROL		0		0		5		0.00	
PHASER_REF		0		0		5		0.00	
OUT_FIFO		0		0		20		0.00	
IN_FIFO		0		0		20		0.00	
IDELAYCTRL		0		0		5		0.00	
IBUFDS		0		0		104		0.00	
GTPE2_CHANNEL		0		0		2		0.00	
PHASER_OUT/PHASER_OUT_PHY		0		0		20		0.00	
PHASER_IN/PHASER_IN_PHY		0		0		20		0.00	
IDELAYE2/IDELAYE2_FINEDELAY		0		0		250		0.00	
IBUFDS_GTE2		0		0		2		0.00	
ILOGIC		0		0		106		0.00	
OLOGIC		0		0		106		0.00	
+-----+-----+-----+-----+-----+									

5. Clocking

+-----+-----+-----+-----+-----+									
Site Type		Used		Fixed		Available		Util%	
+-----+-----+-----+-----+-----+									
BUFGCTRL		1		0		32		3.13	
BUFIO		0		0		20		0.00	
MMCME2_ADV		0		0		5		0.00	
PLLE2_ADV		0		0		5		0.00	
BUFMRCE		0		0		10		0.00	
BUFHCE		0		0		72		0.00	
BUFR		0		0		20		0.00	
+-----+-----+-----+-----+-----+									

6. Specific Feature

+-----+-----+-----+-----+-----+									
Site Type		Used		Fixed		Available		Util%	
+-----+-----+-----+-----+-----+									
BSCANE2		0		0		4		0.00	
CAPTUREE2		0		0		1		0.00	
DNA_PORT		0		0		1		0.00	
EFUSE_USR		0		0		1		0.00	

FRAME_ECCE2	0	0	1	0.00
ICAPE2	0	0	2	0.00
PCIE_2_1	0	0	1	0.00
STARTUPE2	0	0	1	0.00
XADC	0	0	1	0.00
+-----+-----+-----+-----+				

7. Primitives

+-----+-----+-----+		
Ref Name	Used	Functional Category
+-----+-----+-----+		
FDRE	17777	Flop & Latch
IBUF	8265	IO
OBUF	8192	IO
LUT6	7054	LUT
MUXF7	2080	MuxFx
MUXF8	1036	MuxFx
LUT5	463	LUT
LUT2	264	LUT
LUT4	218	LUT
LUT3	173	LUT
CARRY4	114	CarryLogic
FDSE	69	Flop & Latch
LUT1	30	LUT
BUFG	1	Clock
+-----+-----+-----+		

8. Black Boxes

+-----+-----+	
Ref Name	Used
+-----+-----+	

9. Instantiated Netlists

+-----+-----+	
Ref Name	Used

+-----+-----+

ADDITION

POWER REPORT:

Power Report

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 - 3.1 By Hierarchy

1. Summary

+-----+-----+

Total On-Chip Power (W)	140.285 (Junction temp exceeded!)	
Design Power Budget (W)	Unspecified*	
Power Budget Margin (W)	NA	
Dynamic (W)	139.800	
Device Static (W)	0.485	
Effective TJA (C/W)	5.0	
Max Ambient (C)	0.0	
Junction Temperature (C)	125.0	
Confidence Level	Low	
Setting File	---	
Simulation Activity File	---	
Design Nets Matched	NA	

+-----+-----+

* Specify Design Power Budget using, set_operating_conditions -design_power_budget <value in Watts>

1.1 On-Chip Components

On-Chip	Power (W)	Used	Available	Utilization (%)
Slice Logic	13.807	17503	---	---
LUT as Logic	12.732	5833	20800	28.04
F7/F8 Muxes	1.005	3232	32600	9.91
CARRY4	0.058	19	8150	0.23
BUFG	0.006	1	32	3.13
Register	0.005	8364	41600	20.11
Others	0.000	2	---	---
Signals	11.458	27548	---	---
I/O	114.522	24580	106	23188.68
Static Power	0.485			
Total	140.267			

1.2 Power Supply Summary

Source	Voltage (V)	Total (A)	Dynamic (A)	Static (A)
Vccint	1.000	91.659	91.318	0.341
Vccaux	1.800	0.053	0.000	0.053
Vcco33	3.300	0.000	0.000	0.000
Vcco25	2.500	0.000	0.000	0.000
Vcco18	1.800	26.935	26.934	0.001
Vcco15	1.500	0.000	0.000	0.000
Vcco135	1.350	0.000	0.000	0.000
Vcco12	1.200	0.000	0.000	0.000
Vccaux_io	1.800	0.000	0.000	0.000
Vccbram	1.000	0.010	0.000	0.010
MGTAVcc	1.000	0.000	0.000	0.000
MGTAVtt	1.200	0.000	0.000	0.000
Vccadc	1.800	0.020	0.000	0.020

1.3 Confidence Level

+-----+-----+-----+-----+			
-----+			
User Input Data	Confidence	Details	Action
+-----+-----+-----+-----+			
-----+			
Design implementation state Low Design is synthesized Accuracy			
of the tool is not optimal until design is fully placed and routed			
Clock nodes activity Low User specified less than 75% of clocks Provide			
missing clock activity with a constraint file, simulation results or by editing the "By Clock			
Domain" view			
I/O nodes activity Low More than 75% of inputs are missing user specification			
Provide missing input activity with simulation results or by editing the "By Resource Type ->			
I/Os" view			
Internal nodes activity Medium User specified less than 25% of internal nodes			
Provide missing internal nodes activity with simulation results or by editing the "By Resource			
Type" views			
Device models High Device models are Production			
Overall confidence level Low			
+-----+-----+-----+-----+			
-----+			

2. Settings

2.1 Environment

+-----+-----+			
Ambient Temp (C)	25.0		
ThetaJA (C/W)	5.0		
Airflow (LFM)	250		
Heat Sink	medium (Medium Profile)		
ThetaSA (C/W)	4.6		
Board Selection	medium (10"x10")		
# of Board Layers	12to15 (12 to 15 Layers)		
Board Temperature (C)	25.0		
+-----+-----+			

2.2 Clock Constraints

+-----+	+-----+	+-----+
Clock	Domain	Constraint (ns)
+-----+	+-----+	+-----+

3. Detailed Reports

3.1 By Hierarchy

+-----+	+-----+
Name	Power (W)
+-----+	+-----+
target_module	139.783
+-----+	+-----+

0 Infos, 0 Warnings, 0 Critical Warnings and 0 Errors encountered.
report_power completed successfully
report_power: Time (s): cpu = 00:00:42 ; elapsed = 00:00:14 . Memory (MB): peak = 8318.430 ;
gain = 0.000 ; free physical = 293780 ; free virtual = 559672

TIMING REPORT:

Timing Report

Slack: inf
Source: original_matrix[20][27][1]
(input port)
Destination: blended_matrix_reg[0][0][5]/D
Path Group: (none)
Path Type: Max at Slow Process Corner
Data Path Delay: 7.540ns (logic 3.586ns (47.552%) route 3.955ns (52.448%))
Logic Levels: 11 (CARRY4=2 IBUF=1 LUT2=1 LUT5=1 LUT6=3 MUXF7=2 MUXF8=1)

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)

		0.000	0.000 r	original_matrix[20][27][1] (IN)
	net (fo=0)	0.000	0.000	original_matrix[20][27][1]

IBUF (Prop_ibuf_I_O)	0.923	0.923	r
blended_matrix_reg[31][9][3]_i_5938/O			
net (fo=1, unplaced)	0.800	1.722	
blended_matrix_reg[31][9][3]_i_5938_n_0			
LUT6 (Prop_lut6_I5_O)	0.124	1.846	r blended_matrix[31][9][3]_i_2300/O
net (fo=1, unplaced)	0.000	1.846	blended_matrix[31][9][3]_i_2300_n_0
MUXF7 (Prop_muxf7_I1_O)	0.245	2.091	r
blended_matrix_reg[31][9][3]_i_1011/O			
net (fo=1, unplaced)	0.000	2.091	
blended_matrix_reg[31][9][3]_i_1011_n_0			
MUXF8 (Prop_muxf8_I0_O)	0.104	2.195	r
blended_matrix_reg[31][9][3]_i_367/O			
net (fo=1, unplaced)	0.717	2.912	
blended_matrix_reg[31][9][3]_i_367_n_0			
LUT6 (Prop_lut6_I0_O)	0.319	3.231	r blended_matrix[31][9][3]_i_140/O
net (fo=1, unplaced)	0.000	3.231	blended_matrix[31][9][3]_i_140_n_0
MUXF7 (Prop_muxf7_I1_O)	0.245	3.476	r
blended_matrix_reg[31][9][3]_i_59/O			
net (fo=2, unplaced)	0.916	4.392	
blended_matrix_reg[31][9][3]_i_59_n_0			
LUT6 (Prop_lut6_I1_O)	0.298	4.690	r blended_matrix[31][9][3]_i_36/O
net (fo=1, unplaced)	0.902	5.592	blended_matrix[31][9][3]_i_36_n_0
LUT5 (Prop_lut5_I1_O)	0.124	5.716	r blended_matrix[31][9][3]_i_9/O
net (fo=1, unplaced)	0.000	5.716	blended_matrix[31][9][3]_i_9_n_0
CARRY4 (Prop_carry4_S[1]_CO[3])			
	0.550	6.266	r blended_matrix_reg[31][9][3]_i_2/CO[3]
net (fo=1, unplaced)	0.009	6.275	
blended_matrix_reg[31][9][3]_i_2_n_0			
CARRY4 (Prop_carry4_CI_O[1])			
	0.348	6.623	r blended_matrix_reg[31][9][7]_i_9/O[1]
net (fo=1, unplaced)	0.611	7.234	p_1_in[5]
LUT2 (Prop_lut2_I0_O)	0.306	7.540	r blended_matrix[31][9][5]_i_1/O
net (fo=1024, unplaced)	0.000	7.540	blended_matrix[31][9][5]_i_1_n_0
FDRE			r blended_matrix_reg[0][0][5]/D

UTILIZATION REPORT:

Utilization Design Information

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1. Slice Logic

Site Type	Used	Fixed	Available	Util%
Slice LUTs*	5833	0	20800	28.04
LUT as Logic	5833	0	20800	28.04
LUT as Memory	0	0	9600	0.00
Slice Registers	8364	0	41600	20.11
Register as Flip Flop	8364	0	41600	20.11
Register as Latch	0	0	41600	0.00
F7 Muxes	2176	0	16300	13.35
F8 Muxes	1056	0	8150	12.96

* Warning! The Final LUT count, after physical optimizations and full implementation, is typically lower. Run opt_design after synthesis, if not already completed, for a more realistic count.

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	-	-	-
0	-	-	Set
0	-	-	Reset
0	-	Set	-
0	-	Reset	-
0	Yes	-	-
0	Yes	-	Set
172	Yes	-	Reset
0	Yes	Set	-

8192	Yes	Reset	-
+-----+	+-----+	+-----+	+-----+

2. Memory

+-----+	+-----+	+-----+	+-----+	+-----+
Site Type	Used	Fixed	Available	Util%
+-----+	+-----+	+-----+	+-----+	+-----+
Block RAM Tile	0	0	50	0.00
RAMB36/FIFO*	0	0	50	0.00
RAMB18	0	0	100	0.00
+-----+	+-----+	+-----+	+-----+	+-----+

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

3. DSP

+-----+	+-----+	+-----+	+-----+	+-----+
Site Type	Used	Fixed	Available	Util%
+-----+	+-----+	+-----+	+-----+	+-----+
DSPs	0	0	90	0.00
+-----+	+-----+	+-----+	+-----+	+-----+

4. IO and GT Specific

+-----+	+-----+	+-----+	+-----+	+-----+
Site Type	Used	Fixed	Available	Util%
+-----+	+-----+	+-----+	+-----+	+-----+
Bonded IOB	24580	0	106	23188.68
Bonded IPADs	0	0	10	0.00
Bonded OPADs	0	0	4	0.00
PHY_CONTROL	0	0	5	0.00
PHASER_REF	0	0	5	0.00
OUT_FIFO	0	0	20	0.00
IN_FIFO	0	0	20	0.00
IDELAYCTRL	0	0	5	0.00
IBUFDS	0	0	104	0.00

GTPE2_CHANNEL	0	0	2	0.00
PHASER_OUT/PHASER_OUT_PHY	0	0	20	0.00
PHASER_IN/PHASER_IN_PHY	0	0	20	0.00
IDELAYE2/IDELAYE2_FINEDELAY	0	0	250	0.00
IBUFDS_GTE2	0	0	2	0.00
ILOGIC	0	0	106	0.00
OLOGIC	0	0	106	0.00
+-----+-----+-----+-----+				

5. Clocking

+-----+-----+-----+-----+				
Site Type	Used	Fixed	Available	Util%
+-----+-----+-----+-----+				
BUFGCTRL	1	0	32	3.13
BUFIO	0	0	20	0.00
MMCME2_ADV	0	0	5	0.00
PLLE2_ADV	0	0	5	0.00
BUFMRCE	0	0	10	0.00
BUFHCE	0	0	72	0.00
BUFR	0	0	20	0.00
+-----+-----+-----+-----+				

6. Specific Feature

+-----+-----+-----+-----+				
Site Type	Used	Fixed	Available	Util%
+-----+-----+-----+-----+				
BSCANE2	0	0	4	0.00
CAPTUREE2	0	0	1	0.00
DNA_PORT	0	0	1	0.00
EFUSE_USR	0	0	1	0.00
FRAME_ECCE2	0	0	1	0.00
ICAPE2	0	0	2	0.00
PCIE_2_1	0	0	1	0.00
STARTUPE2	0	0	1	0.00
XADC	0	0	1	0.00
+-----+-----+-----+-----+				

7. Primitives

+-----+-----+-----+		
Ref Name	Used	Functional Category
+-----+-----+-----+		
IBUF	16387	IO
OBUF	8193	IO
FDRE	8192	Flop & Latch
LUT6	5069	LUT
MUXF7	2176	MuxFx
MUXF8	1056	MuxFx
LUT5	340	LUT
LUT2	291	LUT
FDCE	172	Flop & Latch
LUT4	111	LUT
LUT1	41	LUT
LUT3	33	LUT
CARRY4	19	CarryLogic
BUFG	1	Clock
+-----+-----+-----+		

8. Black Boxes

+-----+-----+	
Ref Name	Used
+-----+-----+	

9. Instantiated Netlists

+-----+-----+	
Ref Name	Used
+-----+-----+	

NORMALIZATION

POWER REPORT:

Power Report

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1. Summary

```
+-----+-----+
| Total On-Chip Power (W) | 92.869 (Junction temp exceeded!) |
| Design Power Budget (W) | Unspecified* |
| Power Budget Margin (W) | NA |
| Dynamic (W) | 92.385 |
| Device Static (W) | 0.485 |
| Effective TJA (C/W) | 5.0 |
| Max Ambient (C) | 0.0 |
| Junction Temperature (C) | 125.0 |
| Confidence Level | Low |
| Setting File | --- |
| Simulation Activity File | --- |
| Design Nets Matched | NA |
+-----+-----+
```

* Specify Design Power Budget using, set_operating_conditions -design_power_budget <value in Watts>

1.1 On-Chip Components

```
+-----+-----+-----+-----+-----+
```

On-Chip	Power (W)	Used	Available	Utilization (%)
Slice Logic	13.168	14212	---	---
LUT as Logic	11.745	3965	20800	19.06
CARRY4	0.954	115	8150	1.41
F7/F8 Muxes	0.463	1632	32600	5.01
BUFG	0.006	1	32	3.13
Register	<0.001	8317	41600	19.99
Others	0.000	2	---	---
Signals	21.941	18929	---	---
I/O	57.265	16387	106	15459.43
Static Power	0.485			
Total	92.856			

1.2 Power Supply Summary

Source	Voltage (V)	Total (A)	Dynamic (A)	Static (A)
Vccint	1.000	68.484	68.143	0.341
Vccaux	1.800	0.053	0.000	0.053
Vcco33	3.300	0.000	0.000	0.000
Vcco25	2.500	0.000	0.000	0.000
Vcco18	1.800	13.468	13.467	0.001
Vcco15	1.500	0.000	0.000	0.000
Vcco135	1.350	0.000	0.000	0.000
Vcco12	1.200	0.000	0.000	0.000
Vccaux_io	1.800	0.000	0.000	0.000
Vccbram	1.000	0.010	0.000	0.010
MGTAVcc	1.000	0.000	0.000	0.000
MGTAVtt	1.200	0.000	0.000	0.000
Vccadc	1.800	0.020	0.000	0.020

1.3 Confidence Level

User Input Data	Confidence	Details	Action
-----+			
Design implementation state	Low	Design is synthesized	Accuracy
of the tool is not optimal until design is fully placed and routed			
Clock nodes activity	Low	User specified less than 75% of clocks	Provide
missing clock activity with a constraint file, simulation results or by editing the "By Clock Domain" view			
I/O nodes activity	Low	More than 75% of inputs are missing user specification	
Provide missing input activity with simulation results or by editing the "By Resource Type -> I/Os" view			
Internal nodes activity	Medium	User specified less than 25% of internal nodes	
Provide missing internal nodes activity with simulation results or by editing the "By Resource Type" views			
Device models	High	Device models are Production	
Overall confidence level	Low		
-----+			
-----+			

2. Settings

2.1 Environment

-----+			
Ambient Temp (C)	25.0		
ThetaJA (C/W)	5.0		
Airflow (LFM)	250		
Heat Sink	medium (Medium Profile)		
ThetaSA (C/W)	4.6		
Board Selection	medium (10"x10")		
# of Board Layers	12to15 (12 to 15 Layers)		
Board Temperature (C)	25.0		
-----+			

2.2 Clock Constraints

+-----+-----+-----+		
Clock Domain Constraint (ns)		
+-----+-----+-----+		

3. Detailed Reports

3.1 By Hierarchy

+-----+-----+	
Name	Power (W)
+-----+-----+	
target_module	92.372
+-----+-----+	

1 Infos, 1 Warnings, 0 Critical Warnings and 0 Errors encountered.
report_power completed successfully
report_power: Time (s): cpu = 00:03:06 ; elapsed = 00:00:33 . Memory (MB): peak = 8631.832 ;
gain = 0.000 ; free physical = 297160 ; free virtual = 563050

TIMING REPORT:

Timing Report

Slack: inf
Source: input_matrix[20][27][4]
(input port)
Destination: normalized_matrix_reg[0][0][0]/D
Path Group: (none)
Path Type: Max at Slow Process Corner
Data Path Delay: 22.452ns (logic 10.400ns (46.319%) route 12.053ns (53.681%))
Logic Levels: 33 (CARRY4=14 IBUF=1 LUT2=3 LUT3=4 LUT4=1 LUT5=2 LUT6=4
MUXF7=2 MUXF8=2)

Location	Delay type	Incr(ns)	Path(ns)	Netlist Resource(s)

		0.000	0.000 r	input_matrix[20][27][4] (IN)


```

net (fo=0)          0.000  0.000  input_matrix[20][27][4]
IBUF (Prop_ibuf_I_O)  0.923  0.923 r
normalized_matrix_reg[21][10][7]_i_9969/O
net (fo=1, unplaced)  0.800  1.722
normalized_matrix_reg[21][10][7]_i_9969_n_0
LUT6 (Prop_lut6_I5_O)  0.124  1.846 r
normalized_matrix[21][10][7]_i_4539/O
net (fo=1, unplaced)  0.000  1.846
normalized_matrix[21][10][7]_i_4539_n_0
MUXF7 (Prop_muxf7_I1_O)  0.245  2.091 r
normalized_matrix_reg[21][10][7]_i_2098/O
net (fo=1, unplaced)  0.000  2.091
normalized_matrix_reg[21][10][7]_i_2098_n_0
MUXF8 (Prop_muxf8_I0_O)  0.104  2.195 r
normalized_matrix_reg[21][10][7]_i_1134/O
net (fo=1, unplaced)  0.717  2.912
normalized_matrix_reg[21][10][7]_i_1134_n_0
LUT6 (Prop_lut6_I0_O)  0.319  3.231 r
normalized_matrix[21][10][7]_i_667/O
net (fo=1, unplaced)  0.000  3.231
normalized_matrix[21][10][7]_i_667_n_0
MUXF7 (Prop_muxf7_I1_O)  0.245  3.476 r
normalized_matrix_reg[21][10][7]_i_496/O
net (fo=1, unplaced)  0.000  3.476
normalized_matrix_reg[21][10][7]_i_496_n_0
MUXF8 (Prop_muxf8_I0_O)  0.104  3.580 r
normalized_matrix_reg[21][10][7]_i_396/O
net (fo=1, unplaced)  0.717  4.297
normalized_matrix_reg[21][10][7]_i_396_n_0
LUT6 (Prop_lut6_I0_O)  0.319  4.616 r
normalized_matrix[21][10][7]_i_262/O
net (fo=4, unplaced)  1.135  5.751  input_matrix[4]
LUT5 (Prop_lut5_I0_O)  0.124  5.875 r
normalized_matrix[21][10][7]_i_243/O
net (fo=5, unplaced)  0.477  6.352
normalized_matrix[21][10][7]_i_243_n_0
LUT2 (Prop_lut2_I1_O)  0.124  6.476 r
normalized_matrix[21][10][7]_i_246/O
net (fo=8, unplaced)  0.487  6.963
normalized_matrix[21][10][7]_i_246_n_0
LUT2 (Prop_lut2_I1_O)  0.124  7.087 r
normalized_matrix[21][10][7]_i_253/O
net (fo=1, unplaced)  0.000  7.087
normalized_matrix[21][10][7]_i_253_n_0

```

```

CARRY4 (Prop_carry4_S[1]_CO[3])
0.550 7.637 r
normalized_matrix_reg[21][10][7]_i_171/CO[3]
net (fo=1, unplaced) 0.000 7.637
normalized_matrix_reg[21][10][7]_i_171_n_0
CARRY4 (Prop_carry4_CI_CO[3])
0.114 7.751 r normalized_matrix_reg[21][10][7]_i_82/CO[3]
net (fo=29, unplaced) 0.987 8.738
normalized_matrix_reg[21][10][7]_i_82_n_0
LUT2 (Prop_lut2_l1_O) 0.124 8.862 r
normalized_matrix[21][10][7]_i_297/O
net (fo=1, unplaced) 0.449 9.311
normalized_matrix[21][10][7]_i_297_n_0
LUT6 (Prop_lut6_l1_O) 0.124 9.435 r
normalized_matrix[21][10][7]_i_193/O
net (fo=2, unplaced) 0.650 10.085
normalized_matrix[21][10][7]_i_193_n_0
CARRY4 (Prop_carry4_DI[1]_CO[3])
0.507 10.592 r
normalized_matrix_reg[21][10][7]_i_121/CO[3]
net (fo=1, unplaced) 0.000 10.592
normalized_matrix_reg[21][10][7]_i_121_n_0
CARRY4 (Prop_carry4_CI_O[3])
0.329 10.921 r
normalized_matrix_reg[21][10][7]_i_100/O[3]
net (fo=30, unplaced) 0.688 11.609
normalized_matrix_reg[21][10][7]_i_100_n_4
LUT3 (Prop_lut3_l0_O) 0.307 11.916 r
normalized_matrix[21][10][1]_i_135/O
net (fo=1, unplaced) 0.639 12.555
normalized_matrix[21][10][1]_i_135_n_0
CARRY4 (Prop_carry4_DI[1]_CO[3])
0.507 13.062 r
normalized_matrix_reg[21][10][1]_i_90/CO[3]
net (fo=1, unplaced) 0.000 13.062
normalized_matrix_reg[21][10][1]_i_90_n_0
CARRY4 (Prop_carry4_CI_O[2])
0.256 13.318 r normalized_matrix_reg[21][10][5]_i_35/O[2]
net (fo=3, unplaced) 0.440 13.758
normalized_matrix_reg[21][10][5]_i_35_n_5
LUT3 (Prop_lut3_l1_O) 0.301 14.059 r
normalized_matrix[21][10][5]_i_24/O
net (fo=1, unplaced) 0.639 14.698
normalized_matrix[21][10][5]_i_24_n_0

```

```

CARRY4 (Prop_carry4_DI[1]_CO[3])
    0.507  15.205 r
normalized_matrix_reg[21][10][5]_i_12/CO[3]
    net (fo=1, unplaced)    0.000  15.205
normalized_matrix_reg[21][10][5]_i_12_n_0
    CARRY4 (Prop_carry4_CI_O[3])
        0.329  15.534 r normalized_matrix_reg[21][10][7]_i_52/O[3]
        net (fo=3, unplaced)    0.636  16.170
normalized_matrix_reg[21][10][7]_i_52_n_4
    LUT3 (Prop_lut3_I1_O)    0.307  16.477 r
normalized_matrix[21][10][7]_i_22/O
    net (fo=1, unplaced)    0.333  16.810
normalized_matrix[21][10][7]_i_22_n_0
    CARRY4 (Prop_carry4_DI[0]_CO[3])
        0.526  17.336 r normalized_matrix_reg[21][10][7]_i_8/CO[3]
        net (fo=1, unplaced)    0.000  17.336
normalized_matrix_reg[21][10][7]_i_8_n_0
    CARRY4 (Prop_carry4_CI_O[3])
        0.329  17.665 r
normalized_matrix_reg[21][10][7]_i_208/O[3]
    net (fo=9, unplaced)    0.641  18.306
normalized_matrix_reg[21][10][7]_i_208_n_4
    LUT3 (Prop_lut3_I0_O)    0.307  18.613 r
normalized_matrix[21][10][7]_i_129/O
    net (fo=1, unplaced)    0.639  19.252
normalized_matrix[21][10][7]_i_129_n_0
    CARRY4 (Prop_carry4_DI[1]_CO[3])
        0.507  19.759 r
normalized_matrix_reg[21][10][7]_i_54/CO[3]
    net (fo=1, unplaced)    0.000  19.759
normalized_matrix_reg[21][10][7]_i_54_n_0
    CARRY4 (Prop_carry4_CI_O[1])
        0.348  20.107 r normalized_matrix_reg[21][10][7]_i_27/O[1]
        net (fo=3, unplaced)    0.629  20.736
normalized_matrix_reg[21][10][7]_i_27_n_6
    LUT4 (Prop_lut4_I1_O)    0.306  21.042 r
normalized_matrix[21][10][7]_i_41/O
    net (fo=1, unplaced)    0.000  21.042
normalized_matrix[21][10][7]_i_41_n_0
    CARRY4 (Prop_carry4_S[1]_CO[3])
        0.550  21.592 r
normalized_matrix_reg[21][10][7]_i_13/CO[3]
    net (fo=1, unplaced)    0.000  21.592
normalized_matrix_reg[21][10][7]_i_13_n_0

```

```

CARRY4 (Prop_carry4_CI_CO[1])
0.178 21.770 r normalized_matrix_reg[21][10][7]_i_6/CO[1]
net (fo=8, unplaced) 0.350 22.120
normalized_matrix_reg[21][10][7]_i_6_n_2
LUT5 (Prop_lut5_I0_O) 0.332 22.452 r
normalized_matrix[21][10][0]_i_1/O
net (fo=1024, unplaced) 0.000 22.452
normalized_matrix[21][10][0]_i_1_n_0
FDRE r normalized_matrix_reg[0][0][0]/D
-----

```

UTILIZATION REPORT:

Utilization Design Information

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1. Slice Logic

Site Type	Used	Fixed	Available	Util%
Slice LUTs*	3965	0	20800	19.06
LUT as Logic	3965	0	20800	19.06
LUT as Memory	0	0	9600	0.00
Slice Registers	8317	0	41600	19.99
Register as Flip Flop	8317	0	41600	19.99
Register as Latch	0	0	41600	0.00
F7 Muxes	1088	0	16300	6.67
F8 Muxes	544	0	8150	6.67

* Warning! The Final LUT count, after physical optimizations and full implementation, is typically lower. Run opt_design after synthesis, if not already completed, for a more realistic count.

1.1 Summary of Registers by Type

	Total	Clock Enable	Synchronous	Asynchronous
0	_	-	-	
0	_	-	Set	
0	_	-	Reset	
0	_	Set	-	
0	_	Reset	-	
0	Yes	-	-	
0	Yes	-	Set	
125	Yes	-	Reset	
0	Yes	Set	-	
8192	Yes	Reset	-	

2. Memory

Site Type	Used	Fixed	Available	Util%
Block RAM Tile	0	0	50	0.00
RAMB36/FIFO*	0	0	50	0.00
RAMB18	0	0	100	0.00

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

3. DSP

Site Type	Used	Fixed	Available	Util%
-----------	------	-------	-----------	-------

DSPs	0	0	90	0.00
+-----+-----+-----+-----+				

4. IO and GT Specific

Site Type	Used	Fixed	Available	Util%
Bonded IOB	16387	0	106	15459.43
Bonded IPADs	0	0	10	0.00
Bonded OPADs	0	0	4	0.00
PHY_CONTROL	0	0	5	0.00
PHASER_REF	0	0	5	0.00
OUT_FIFO	0	0	20	0.00
IN_FIFO	0	0	20	0.00
IDELAYCTRL	0	0	5	0.00
IBUFDS	0	0	104	0.00
GTPE2_CHANNEL	0	0	2	0.00
PHASER_OUT/PHASER_OUT_PHY	0	0	20	0.00
PHASER_IN/PHASER_IN_PHY	0	0	20	0.00
IDELAYE2/IDELAYE2_FINEDELAY	0	0	250	0.00
IBUFDS_GTE2	0	0	2	0.00
ILOGIC	0	0	106	0.00
OLOGIC	0	0	106	0.00

5. Clocking

Site Type	Used	Fixed	Available	Util%
BUFGCTRL	1	0	32	3.13
BUFIO	0	0	20	0.00
MMCM2_ADV	0	0	5	0.00
PLLE2_ADV	0	0	5	0.00
BUFMRCE	0	0	10	0.00
BUFHCE	0	0	72	0.00
BUFR	0	0	20	0.00

6. Specific Feature

Site Type	Used	Fixed	Available	Util%
BSCANE2	0	0	4	0.00
CAPTUREE2	0	0	1	0.00
DNA_PORT	0	0	1	0.00
EFUSE_USR	0	0	1	0.00
FRAME_ECCE2	0	0	1	0.00
ICAPE2	0	0	2	0.00
PCIE_2_1	0	0	1	0.00
STARTUPE2	0	0	1	0.00
XADC	0	0	1	0.00

7. Primitives

Ref Name	Used	Functional Category
IBUF	8194	IO
OBUF	8193	IO
FDRE	8192	Flop & Latch
LUT6	2813	LUT
MUXF7	1088	MuxFx
LUT5	631	LUT
MUXF8	544	MuxFx
LUT2	258	LUT
LUT3	236	LUT
LUT4	180	LUT
FDCE	125	Flop & Latch
CARRY4	115	CarryLogic
LUT1	27	LUT
BUFG	1	Clock

8. Black Boxes

```

+-----+-----+
| Ref Name | Used |
+-----+-----+

```

9. Instantiated Netlists

```

+-----+-----+
| Ref Name | Used |
+-----+-----+

```

Table comparing all modules:

Module	Power(W)	Delay(ns)	LUTs
Smoothing	95.871	15.258	10825
Sharpening	96.465	12.771	8053
Blending/Addition	140.267	7.54	5833
Normalization	92.856	22.452	3965

Python code for finding minimum and maximum pixel value

```

def find_min_max_in_hex_file(file_path):
    """
    Find the minimum and maximum integer values in a file containing hex
    values.

    Parameters:
    - file_path: Path to the text file containing hex values, one per line
    or space-separated.

    Returns:
    - A tuple containing the minimum and maximum values.
    """
    min_val = float('inf')
    max_val = float('-inf')

```



```
with open(file_path, 'r') as file:
    for line in file:

        hex_values = line.strip().split()
        for hex_value in hex_values:
            # Convert hex to int
            int_value = int(hex_value, 16)

            min_val = min(min_val, int_value)
            max_val = max(max_val, int_value)

        return min_val, max_val

file_path = 'blended_image.txt'
min_val, max_val = find_min_max_in_hex_file(file_path)
print(f"Minimum value: {min_val}, Maximum value: {max_val}")
```

Result Images



sharpen



smooth



blended



normalized

How we will pipeline the system:

We will do pipeline of the stages between smoothing and sharpening. So, that when one image is the smoothing stage then previous image can be in the sharpening stage. Addition of

Instantiate and connect the stages of the pipeline

```
gaussian_blur #(.ROWS(IMG_HEIGHT), .COLS(IMG_WIDTH)) gaussian_blur_stage (  
    .clk(clk),  
    .reset(reset),  
    .enable(enable),  
    .image_in(image_in),  
    .image_out(smoothed_image),  
    .done(smoothing_done)  
);
```

```

sharpening_stage #(.ROWS(IMG_HEIGHT), .COLS(IMG_WIDTH)) sharpening_stage (
    .clk(clk),
    .reset(reset),
    .enable(smoothing_done), // Triggered after smoothing is done
    .image_in(smoothed_image),
    .image_out(sharpened_image),
    .done(sharpening_done)
);

adding_stage #(.ROWS(IMG_HEIGHT), .COLS(IMG_WIDTH)) adding_stage (
    .clk(clk),
    .reset(reset),
    .enable(sharpening_done), // Triggered after sharpening is done
    .original_image(image_in),
    .sharpened_image(sharpened_image),
    .image_out(added_image),
    .done(adding_done)
);

normalization_stage #(.ROWS(IMG_HEIGHT), .COLS(IMG_WIDTH)) normalization_stage (
    .clk(clk),
    .reset(reset),
    .enable(adding_done), // Triggered after adding is done
    .image_in(added_image),
    .image_out(image_out),
    .done(done) // Final output
);

endmodule

```